
Vehical service
Management system
Vehicle Service Management System

Project Overview:

The Vehicle Service Management System is a web-based application designed to streamline and automate the process of vehicle maintenance and repair services. It enables users to book services, track service history, receive automated reminders, and communicate with service centers efficiently.

The platform supports admin, customer, and mechanic roles, ensuring smooth coordination, reducing paperwork, and improving customer satisfaction through timely updates and transparent service records.





Problem Statement:

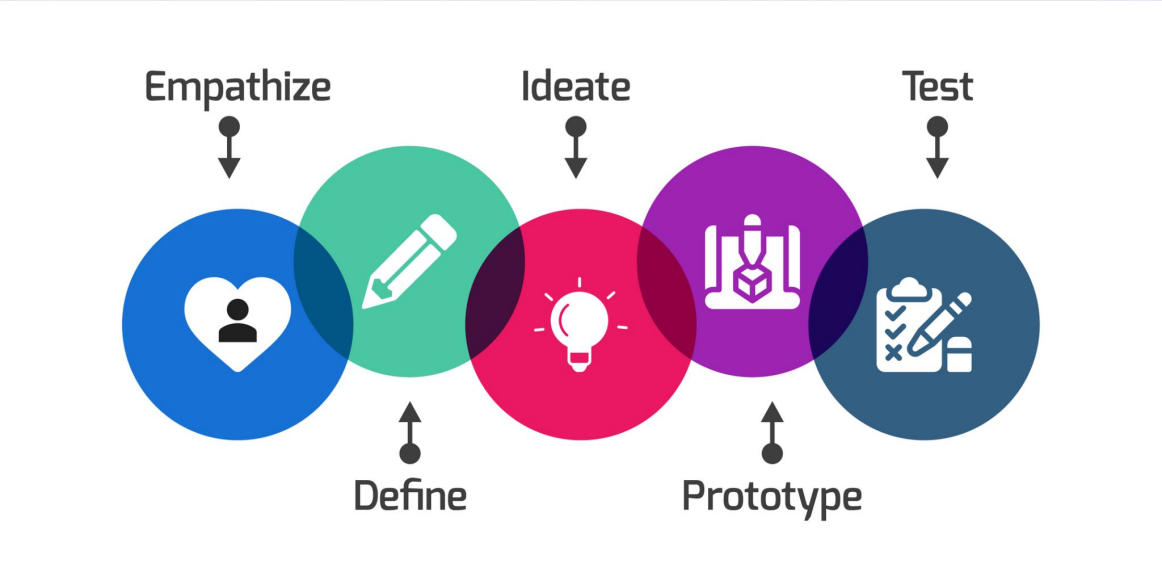
Many vehicle owners forget or delay regular maintenance due to the absence of timely reminders and organized service tracking. The traditional process of visiting or calling service centers is time-consuming, inefficient, and lacks transparency. Customers often face issues like poor service communication, unclear billing, and no way to view past service records.



Goal

The goal of the Vehicle Service Management System is to digitize and automate the entire vehicle service lifecycle. It aims to create a user-friendly platform where vehicle owners can easily book services, get automated reminders, and track service status in real-time. It will maintain digital records of past services, send notifications for upcoming maintenance, and allow smooth interaction between customers and service centers.

Design Thinking



1 **Empathize:**

- **User Research:**Conducted surveys with mechanics and service advisors at local garages. Spoke with daily commuters, taxi drivers, and private car owners.

- **Observation:** Forget service dates and face engine issues due to missed maintenance. Experience long wait times and poor communication with service centers.

2. Define:

- **Problem Statement:** Vehicle owners often face challenges in remembering service dates, booking appointments conveniently, and staying informed about the status of their vehicle during servicing. The lack of timely reminders, transparency, and clear communication leads to missed services, vehicle issues, and frustration
- **Persona:**Private car owners who need regular maintenance for personal or family vehicles.
- People with busy schedules who prefer online solutions over in-person booking.

3. Ideate:

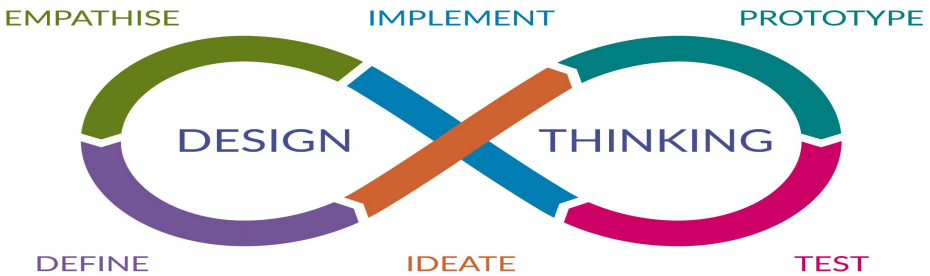
- **Brainstorm Solutions:** An easy-to-use website or mobile app for customers to book services online anytime.
- Automated reminders through SMS or email to notify customers about upcoming services or insurance renewals.
- A live service tracker so customers can see each stage of their vehicle's servicing in real time.
- **Sketch Ideas:** Draft wireframes showing dashboards for user flow and make design easy to use for boyh customers and staff

4. Prototype:

- Low-Fidelity Prototype: Develop prototypes included screens for booking services, tracking service progress, viewing service history, and the admin dashboard.
- Iterate: the design by improving the layout, simplifying navigation, and adding clearer labels to make the system more user-friendly.

5. Test:

- Usability Testing: We conducted usability testing with a few vehicle owners and mechanics to observe how easily they could book a service, check service status, and view service history using the prototype.
- Refine: Improve the app design based on feedback, ensuring the user experience is smooth and efficient.



Vehicle service management system

Understanding the user

01 *User Research*

02 *Personas*

03 *Problem Statement*

04 *User Journey Map*

🔍 User Research Summary

To understand the needs and challenges of both vehicle owners and service centers, we conducted interviews, surveys, and observations with more than 15 participants, including daily commuters, car owners, mechanics, and service advisors.

Vehicle Owners: Forget service or insurance renewal dates.Face long wait times and unclear service status updates.Have difficulty maintaining or accessing past service records.

Service Centers: Struggle with managing many bookings manually.Have limited ways to keep customers informed about service progress.Deal with errors in recording service details, leading to customer complaints.

Customers want a convenient platform to book services and get reminders.Both customers and staff need real-time updates to improve communication.

PERSON:PRIYA

Priya wants a simple, reliable app that saves her time, helps her plan ahead, and ensures her car stays in good condition.

Problem Statement:

Problem Statement: priya struggles to remember her car's service dates due to her busy work and family schedule. She finds it inconvenient to call service centers or wait in long queues.



Age : 23
Education:Btech
Hometown:Delhi
Family:Lives with family
Occupation:Software Engineer

🎯 Goals:

- Book car services online anytime.
- Receive reminders so she doesn't miss services.
- Track her car's service progress in real time.
- Keep a digital record of her vehicle's service history

😠 Frustrations:

- Wasting time on rep calls to service cent
- Forgetting important or insurance dates.
- Not knowing exactly the car will be ready

PERSON:RAJESH

Rajesh wants a flexible system he can use on the go to book, reschedule, and stay informed about his car's maintenance without hassle.

Problem Statement:

Rajesh frequently travels for work and often misses his car's scheduled services. He finds it difficult to coordinate with service centers due to his unpredictable schedule.



Age : 42
Education:MBA
Hometown:Bengaluru
Family: Married,2 children
Occupation: Sales Manager

🎯 Goals:

- Easily reschedule services if plans change.
- Get quick updates on his car's service status.
- View past service records whenever needed.
- Ensure his car is always road-ready for work trips.

😠 Frustrations:

- Lack of flexibility to adjust service appointments.
- Unclear or delayed communication from service centers.
- No central place to track all his vehicle's past services.

Empathy Map

"When is my next car service due?"

"Why don't they update me about my car's status?"

"I wish there was an easier way to book a service."

"I always lose my service receipts."

Says

Frustrated about long waits and unclear updates.

Worried about missing services or insurance renewals.

Relieved when there's clear communication and easy booking.



"I hope the service center doesn't overcharge me."

"It's so difficult to keep track of my vehicle's history."

"What if I forget to service my car on time?"

Thinks

Calls the service center multiple times for updates.

Tries to keep physical receipts but often misplaces them.

Delays or skips services due to busy schedules.

Searches online for trusted service options but finds it confusing.

User Research : Pain points

1

🚗Customer :

Forgetting vehicle service or insurance renewal dates.

Difficulty finding a reliable service center nearbyNo option to book services online quickly and easily.Long wait times at service centers without updates. Repeatedly calling to check the service status. Lack of transparency in pricing or service charges Losing paper service records or bills.

2

🏢Service Center:

Managing large numbers of customer bookings manually.Miscommunication or unclear service requests from customers. Difficulty organizing schedules during peak hours. assignments technicians.Delayed service updates cause customer complaints. Difficulty tracking service records for returning customers.

3

👩‍🔧Technician:

Unclear instructions about customer requirements.No proper tool to track assigned jobs and deadlines.Difficulty updating service status in real time.Confusion during peak hours due to poor job scheduling.Manual record-keeping wastes time and causes errors.Lack of coordination with front-desk staff.

GAIN POINTS

1

🛡️Automobile Insurance Providers :

Get access to accurate, up-to-date service records for faster claim processing.Reduce fraud by verifying vehicle maintenance history digitally.Receive reminders about expiring insurance to offer timely renewals.Analyze patterns in vehicle maintenance for better risk assessment.

2

🚘Car Dealerships:

Offer customers seamless service scheduling after purchase.Maintain customer relationships through timely service reminders. Monitor vehicles sold for consistent maintenance, increasing resale value.Use digital records to offer certified pre-owned cars with verified history.

3

🚚 3. Fleet Management Companies:

Monitor servicing schedules for e vehicle in a large fleet automatical Reduce downtime by planning maintenance proactively. Keep accurate digital records of every service for compliance.Receive instant alerts when a vehicle needs attention.Assign vehicles based on maintenance status to avoid breakdowns.

User journey map

The mapping of rajesh user journey Rajesh a simple way to book and manage his car's services . He wants to quickly schedule appointments,updates on service status, and keep a digital record of all his car's past maintenance.

Persona: Rajesh Kumar, 42

Goal: To ensure timely, hassle-free vehicle servicing while balancing a busy travel schedule.

ACTION	Book a Service Conveniently	Receive Service Updates	Track Service History	Adjust Plans if Needed	Review & Provide Feedback
TASK LIST	1. Open app/web portal. 2. Select service center, date, and time.	1. Get live notifications on vehicle status. 2. Contact center if needed.	1. Access list of past services. 2. Check details of work done	1.Reschedule booking if travel plans change. 2. Receive confirmation.	1. Rate service experience. 2. Submit feedback on improvements.
FEELING ADJECTIVE	Relieved to avoid phone calls and queues.	Confident knowing his car's exact service stage.	Satisfied to have clear records.	Calm about flexibility during schedule changes.	Motivated to share thoughts to improve service.
IMPROVEMENT OPPORTUNITIES	1. Provide one-click rebooking. 2. Offer preferred center suggestions based on location.	1. Add ETA updates for each service stage. 2. Enable direct chat with service staff.	1. Include downloadable service reports 2. Highlight upcoming recommended services.	1. Add calendar integration for reminders. 2. Allow quick changes with minimal steps.	1. Offer rewards or discounts for submitting reviews. 2. Summarize feedback trends for service centers.

Vehicle service management system

Starting the Design

01 *Paper Wireframes*

02 *Digital Wireframes*

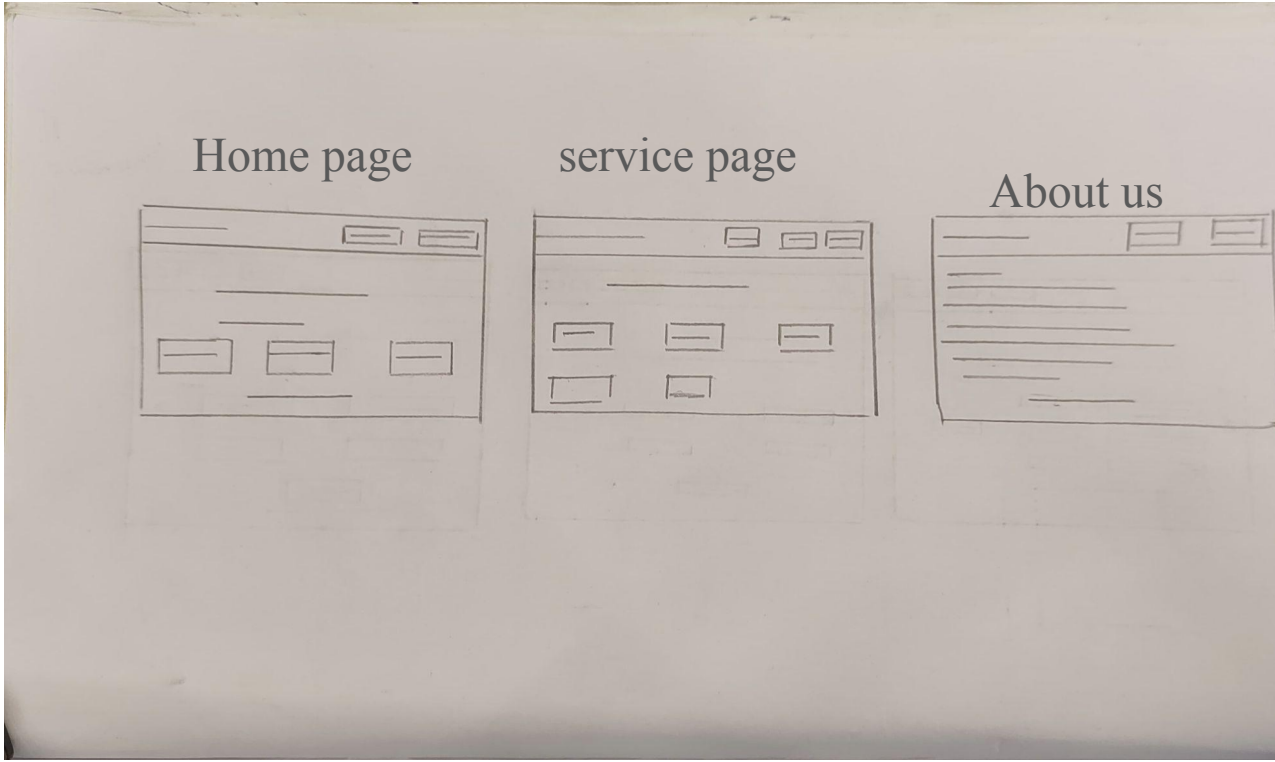
03 *Low Fidelity prototypes*

04 *Usability Studies*

Paper wireframes

Homepage

- Layout:
- Top Bar: App logo on the left, profile icon on the right for easy access to user account.
- 2. Main Buttons: Prominent options for “Book Service,” “Track Service,” and “Service History” centered on the screen.
- 3. Quick Info Section: Upcoming service reminders or notifications displayed below the main buttons.
- 4. Bottom Navigation: Simple menu with icons for Home, Bookings, Notifications, and Settings for easy navigation



About Us

🏢Layout:

1. Header Section: App logo with a page title “About Us” and a brief tagline about trusted vehicle servicing.

2. Company Overview: A section with text describing the mission, values, and benefits of using the system, plus photos of the service team or center.

3. Contact Info & Social Links: Clear display of phone number, email, address, and icons linking to social media pages.

Service Page

🛠️ Service :

1. Service Options Grid/List: Display available services like oil change, brake check, tire rotation, etc., each with icons and short descriptions.

2. Service Details Section: When a service is selected, show expanded information with estimated time, cost, and what’s included.

3. Booking Button: Prominent “Book Now” button below service details to quickly start the booking process.

Book Service

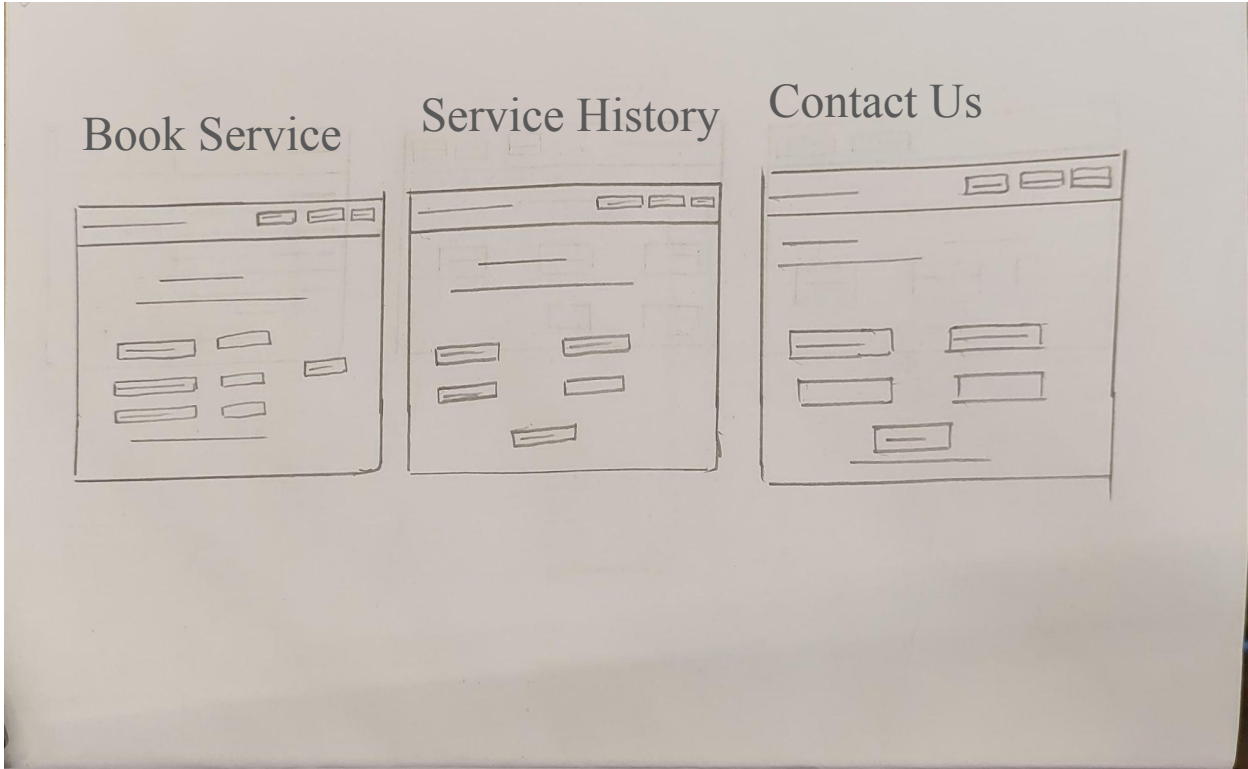
- **Layout:**
  - 1. Booking Form: Dropdowns or fields to select vehicle, service type, preferred date, and time slot.
  - 2. Location Picker: Option to choose or confirm the nearest service center..

Service History

- Layout:
  - 1. History List: Timeline or list view showing past services with dates, service details, and cost.
  - 2. Service Details: Tap or click on any entry to see a summary of the work done, parts replaced, and mechanic notes.
  - 3. Download/Share Option: Button to download service records as PDF or share with insurance/third parties

Contact Us

- Layout:
  - Form with standard fields such as "Name," "Email," and "Message."
  - Larger textarea for the message.
  - Quick contact info (phone, email, address) visible on the side or top right.



DIGITAL WIREFRAMES

Key Screens: Low-fidelity layouts for Home, Book Service, Track Service, Service History, and Admin Dashboard, showing placement of buttons, forms, and navigation.
Interactive Flow: Clickable elements like “Book Now” and “Track Status” demonstrate how users move through booking and tracking processes.

Home page



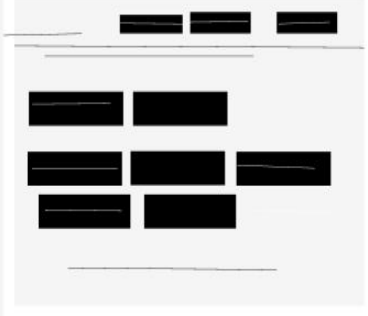
Service Booking



About us



Service History

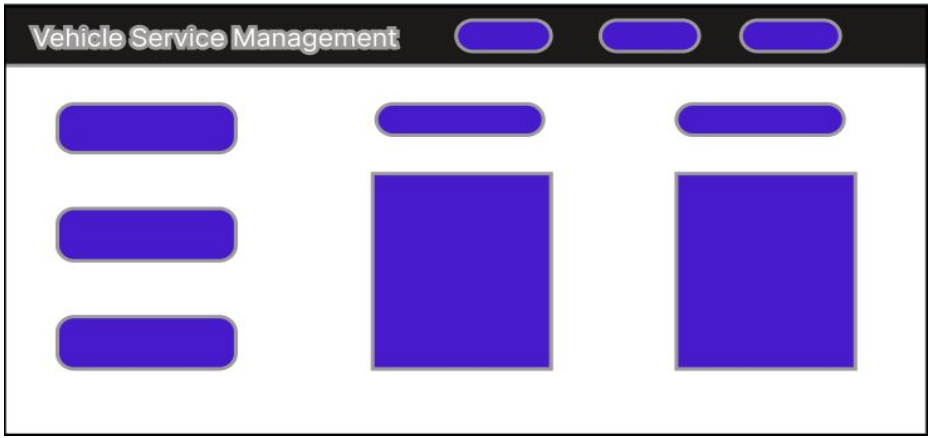


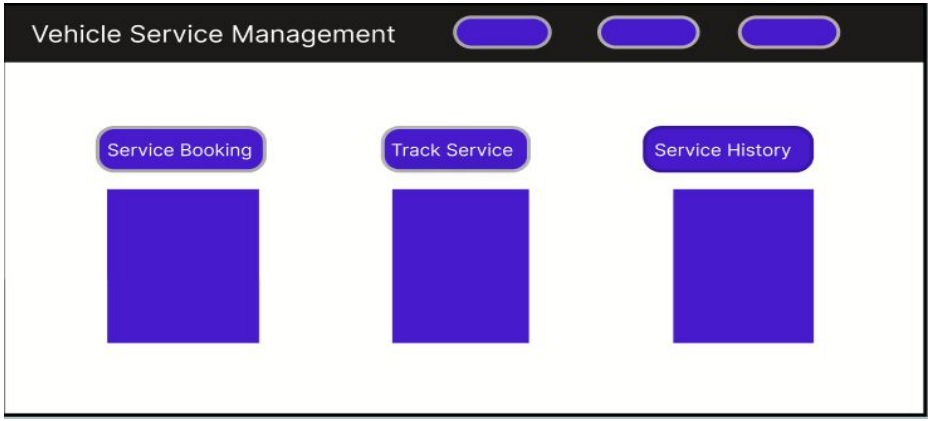




LOW FIDELITY PROTOTYPE

- Basic black-and-white wireframes designed to show layout and user flow without colors, images, or detailed styles.
- Key Screens:Includes Home page with main options, Booking page with form fields, Service Tracker with progress steps, and Service History showing past services.
- Purpose:Quickly test navigation, placement of elements, and overall usability before building high-fidelity designs or coding the app.





Usability Study:

1..Objective:

To evaluate how easily users can navigate the system, complete key tasks (like booking a service), and understand service updates.

2.Participants:

5–7 vehicle owners of different ages and professions who regularly use car services.

3.Method:

Users were asked to complete tasks using the low-fidelity prototype, including booking a service, checking service history, and tracking progress. Observations and interviews captured their behavior and feedback.

4.Key Findings:

Most users found the booking process intuitive but wanted clearer confirmation messages.Users appreciated the simple layout but suggested adding more labels/icons for clarity.

These points highlight key areas for potential improvement based on user feedback, emphasizing navigation, timely information, and community interaction.

COMPETITIVE ANALYSIS

Competitor	Type	Location	Offering	Price	Website	Size	Audience	Unique Value	Impressions
GoMechanic	Direct	India	Car Servicing	Paid	gomechanic.in	Large	Car owners	Doorstep service,Warranty options	Well-known,tech integrated
CarDekho Service	indirect	India	Multibrand car service	Paid	Cardekho.com	Medium	Car buyers/owners	Partner garages,convenience	Trusted brand,growing network
VehicleCare	Direct	India	Car servicing ,diagnostics	Paid	vehiclecare.in	Small	Tech-savvy users	AI-based diagnostics	Modern feel,limted coverage

Competitor	Features	Accessibility	User Flow	Navigation	Brand Identity	Tone	Descriptiveness
GoMechanic	Scheduled service,repair,doorstep	Easy app and web access	Simple booking system	Clear menus,smart filters	Urban,reliable	Professional	Detailed,includes pricing
CarDekho Service	Multi-brand support,comparision	Good web platform,mobile responsive	Guided service	Simple and familiar	Informative,neutral	Friendly	Moderate;visuals and text tips
VehicleCare	AI diagnostics,smart suggestions	App-focused; may lack for all devices	Tech-Based,fast booking	Sleek UI,may needdance gui	innovative,youthful	Modern,direct	Moderate;depends on service type

Vehicle service management system

Refine The Design

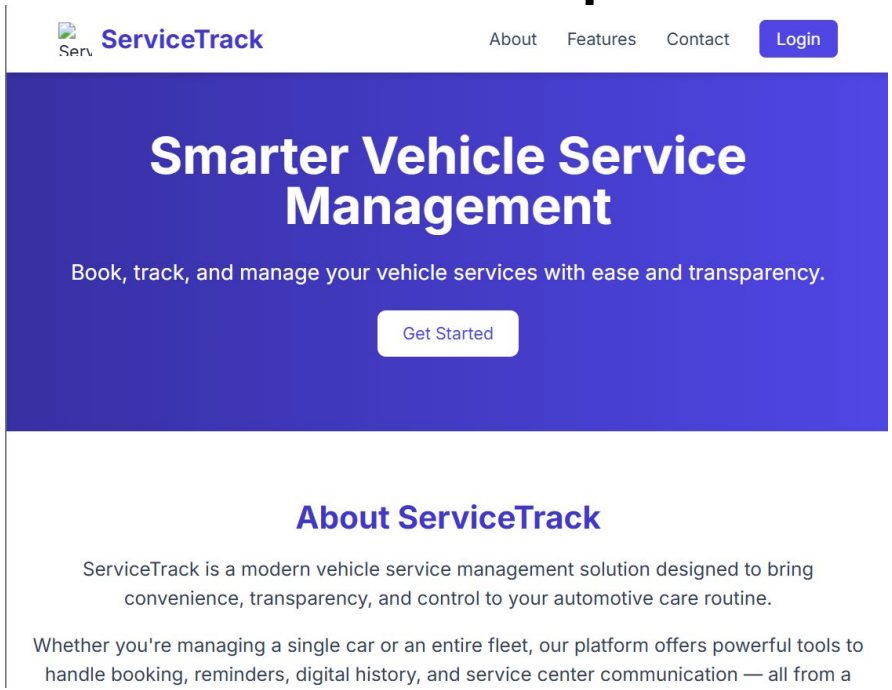
01 *Mockups*

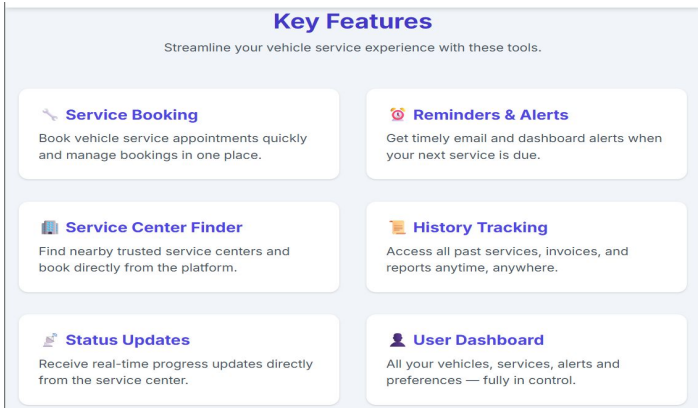
02 *High Fidelity prototypes*

03 *Accessibility*

04 *Workflow*

MOCKUPS





Designs allowed for some iterations, after the usability studies, I added some features in the task bar and I added a more vibrant background for the web page I also revised the design colors

HIGH FIDELITY PROTOTYPE

Welcome, Sravya!

Here's your vehicle service summary and quick tools.

My Vehicles

Swift Dzire – KA01AB1234 (Petrol)
i20 Elite – TS05XY5678 (Diesel)

+ Add Vehicle

Next Due Service

Swift Dzire - **Oil Change**
Due on: 2025-07-10
View Service Details

Upcoming Bookings

July 6, 10:00 AM - AutoFix Center, Hyderabad
Service: AC Checkup
View Booking

Alerts & Reminders

Service History Tracking

View and manage your vehicle's complete service records.

Select Vehicle

Booking ID

Status

Service Type

Service Records

Date	Center	Service	Cost	Status	Actions
2025-07-01	AutoFix, Hyderabad	Oil Change	₹1,200	Completed	View Rebook

Live Service Status Updates

Track your vehicle's service journey in real-time.

Booking Information

Booking ID: #45678
Vehicle: Honda Amaze - TS10AB1234
Service Center: SpeedAuto, Hyderabad
Contact: +91 9876543210
Scheduled: 2025-07-05, 10:00 AM

Service Progress

- **Vehicle Received**
  2025-07-05 10:05 AM
- **Diagnosis Completed**
  2025-07-05 10:30 AM
- **Service In Progress**
  Estimated Completion: 12:30 PM

Find a Service Center

Search and book from trusted vehicle service centers near you.

City, Area, or Pincode

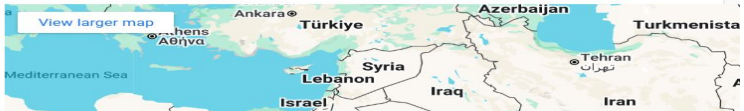
Filter by Rating

Service Type

Auto Detect Location

Sort by: Nearest First

Map View



Reminders & Alerts

Stay up to date with upcoming services, overdue maintenance, and custom notifications.

Upcoming Services

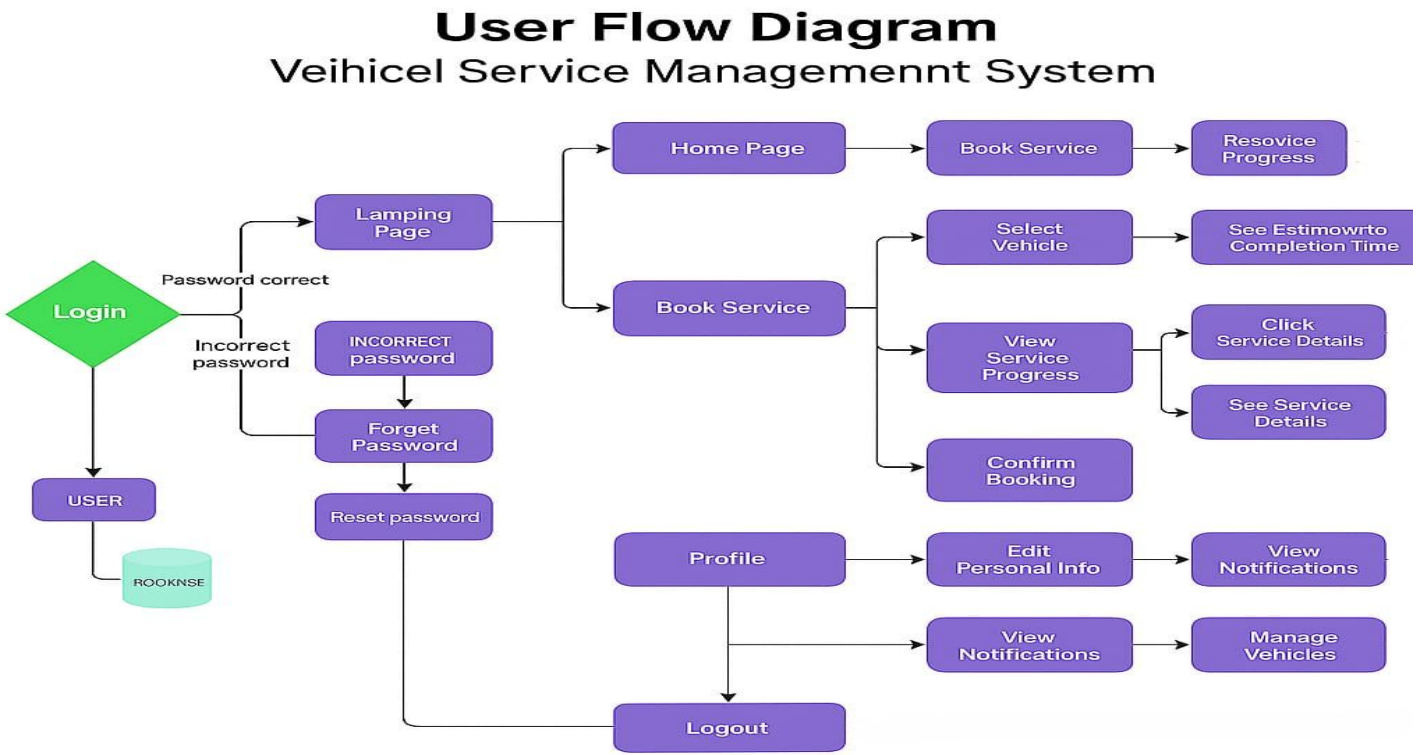
2-tier - Oil Change
Due: 2025-07-02
Upcoming

3-tier - Brake Inspection
Due: 2025-07-05
Upcoming

Overdue Services

4-tier - Battery Check
Due: 2025-06-15
Overdue

USER FLOW DIAGRAM



Vehicle service management system

Going Forward

01 Takeaways

02 Next Types

Next Steps:

1.Enhance Platform Features:
Add advanced options like push notifications, real-time service tracking, and secure online payments.

2.Improve User Experience:
Refine the UI/UX design based on usability testing feedback for smoother booking and navigation.

3. Integrate Data Analytics:
Set up dashboards to track service trends, user behavior, and maintenance patterns for continuous improvement.

4.Expand Service Network:
Partner with more service centers to provide wider coverage and convenience for users.

IMPACT AND TAKEAWAYS

Impact :

Simplifies vehicle service booking, saving customers time and effort.Sends automated reminders for upcoming services and insurance renewals.Provides real-time service tracking, building trust and transparency.Digitizes service history, reducing paperwork and lost records.

Takeaways

A user-friendly platform encourages regular maintenance, extending vehicle lifespan.Clear updates and reduced wait times boost overall customer satisfaction.Data insights enable continuous improvement for both the platform and service centers.Strengthens long-term relationships between vehicle owners and service providers.


Thank
you